Power Question 2013: Power of (Urban) Planning

In ARMLopolis, every house number is a positive integer, and City Hall's address is 0. However, due to the curved nature of the cowpaths that eventually became the streets of ARMLopolis, the distance $d(n)$ between house n and City Hall is not simply the value of n. Instead, if $n = 3^k n'$, where $k \geq 0$ is an integer and n' is an integer not divisible by 3, then $d(n) = 3^{-k}$. For example, $d(18) = 1/9$ and $d(17) = 1$. Notice that even though no houses have negative numbers, $d(n)$ is well-defined for negative values of n. For example, $d(-33) = 1/3$ because $-33 = 3^1 \cdot -11$. By definition, $d(0) = 0$. Following the dictum "location, location, location," this Power Question will refer to "houses" and "house numbers" interchangeably.

Curiously, the arrangement of the houses is such that the distance from house n to house m, written $d(m, n)$, is simply $d(m - n)$. For example, $d(3, 4) = d(-1) = 1$ because $-1 = 3^0 \cdot -1$. In particular, if $m = n$, then $d(m, n) = 0$.

1a. Compute $d(6)$, $d(16)$, and $d(72)$. [1 pt]

1b. Of the houses with positive numbers less than 100, find, with proof, the house or houses which is (are) closest to City Hall. [2 pts]

1c. Find four houses at a distance of $1/9$ from house number 17. [1 pt]

1d. Find an infinite sequence of houses $\{h_n\}$ such that
$$d(17, h_1) > d(17, h_2) > \cdots > d(17, h_n) > d(17, h_{n+1}) > \cdots.$$ [2 pts]

The *neighborhood* of a house n, written $\mathcal{N}(n)$, is the set of all houses that are the same distance from City Hall as n. In symbols, $\mathcal{N}(n) = \{m \mid d(m) = d(n)\}$. Geometrically, it may be helpful to think of $\mathcal{N}(n)$ as a circle centered at City Hall with radius $d(n)$.

2a. Find four houses in $\mathcal{N}(6)$. [1 pt]

2b. Suppose that n is a house with $d(n) = 1$. Find four houses in $\mathcal{N}(n)$. [1 pt]

2c. Suppose that n is a house with $d(n) = 1/27$. Determine the ten smallest positive integers m (in the standard ordering of the integers) such that $m \in \mathcal{N}(n)$. [2 pts]

3a. Notice that $d(16) = d(17) = 1$ and that $d(16, 17) = 1$. Is it true that, for all m, n such that $d(m) = d(n) = 1$ and $m \neq n$, $d(m, n) = 1$? Either prove your answer or find a counterexample. [1 pt]

3b. Suppose that $d(m) = d(n) = 1/3^k$ and that $m \neq n$. Determine the possible values of $d(m, n)$. [2 pts]

3c. Suppose that $d(17, m) = 1/81$. Determine the possible values of $d(16, m)$. [2 pts]

A visitor to ARMLopolis is surprised by the ARMLopolitan distance formula, and starts to wonder about ARMLopolitan geometry. An *ARMLopolitan triangle* is a triangle, all of whose vertices are ARMLopolitan houses.

4a. Show that $d(17, 51) \leq d(17, 34) + d(34, 51)$ and that $d(17, 95) \leq d(17, 68) + d(68, 95)$. [1 pt]

4b. If $a = 17$ and $b = 68$, determine all values of c such that $d(a, c) = d(a, b)$. [2 pts]

5a. Prove that, for all a and b, $d(a, b) \leq \max\{d(a), d(b)\}$. [2 pts]

5b. Prove that, for all a, b, and c, $d(a, c) \leq \max\{d(a, b), d(b, c)\}$. [2 pts]

5c. Prove that, for all a, b, and c, $d(a, c) \leq d(a, b) + d(b, c)$. [1 pt]

5d. After thinking about it some more, the visitor announces that all ARMLopolitan triangles have a special property. What is it? Justify your answer. [2 pts]

Unfortunately for new development, ARMLopolis is full: every nonnegative integer corresponds to (exactly one) house (or City Hall, in the case of 0). However, eighteen families arrive and are looking to move in. After much debate, the connotations of using negative house numbers are deemed unacceptable, and the city decides on an alternative plan. On July 17, Shewad Movers arrive and relocate every family from house n to house $n + 18$, for all positive n (so that City Hall does not move). For example, the family in house number 17 moves to house number 35.

6a. Find at least one house whose distance from City Hall changes as a result of the move. [1 pt]

6b. Prove that the distance between houses with consecutive numbers does not change after the move. [2 pts]

7a. The residents of $\mathcal{N}(1)$ value their suburban location and protest the move, claiming it will make them closer to City Hall. Will it? Justify your answer. [2 pts]

7b. Some residents of $\mathcal{N}(9)$ claim that their tightly-knit community will be disrupted by the move: they will be scattered between different neighborhoods after the change. Will they? Justify your answer. [2 pts]

7c. Other residents of $\mathcal{N}(9)$ claim that their community will be disrupted by newcomers: they say that after the move, their new neighborhood will also contain residents previously from several different old neighborhoods (not just the new arrivals to ARMLopolis). Will it? Justify your answer. [2 pts]

7d. Determine all values of n such that $\mathcal{N}(n)$ will either entirely relocate (i.e., all residents r of $\mathcal{N}(n)$ are at a different distance from City Hall than they were before), lose residents (as in 7b), or gain residents besides the new ARMLopolitans (as in 7c). [2 pts]

8a. One day, Paul (house 23) and Sally (house 32), longtime residents of $\mathcal{N}(1)$, are discussing the "2 side" of the neighborhood, that is, the set of houses $\{n \mid n = 3k + 2, k \in \mathbb{Z}\}$. Paul says "I feel like I'm at the center of the "2 side": when I looked out my window, I realized that the "2 side" consists of exactly those houses whose distance from me is at most 1/3." Prove that Paul is correct. [2 pts]

8b. Sally replies "It's not all about you: I have the same experience!" Justify Sally's claim; in other words, prove that $\{n \mid n = 3k + 2, k \in \mathbb{Z}\}$ consists exactly of those houses whose distance from Sally's house is at most 1/3. [2 pts]

8c. Paul's and Sally's observations can be generalized. For any x, let $\mathcal{D}_r(x) = \{y \mid d(x, y) \leq r\}$, that is, $\mathcal{D}_r(x)$ is the *disk* of radius r. Prove that if $d(x, z) = r$, then $\mathcal{D}_r(x) = \mathcal{D}_r(z)$. [2 pts]

Paul's and Sally's experiences may seem incredible to a newcomer to ARMLopolis. Given a circle, any point on the circle is the center of the circle (so really *a* center of the circle). But ARMLopolitan geometry is even stranger than that!

9a. Ross takes a walk starting at his house, which is number 34. He first visits house n_1, such that $d(n_1, 34) = 1/3$. He then goes to another house, n_2, such that $d(n_1, n_2) = 1/3$. Continuing in that way, he visits houses $n_3, n_4, \ldots$, and each time, $d(n_i, n_{i+1}) = 1/3$. At the end of the day, what is his maximum possible distance from his original house? Justify your answer. [2 pts]

9b. Generalize your answer to 9a: given a value of n_0, determine the rational values of r for which the sequence $n_1, n_2, \ldots$ with $d(n_i, n_{i+1}) = r$ is entirely contained in a circle of finite radius centered at n_0. [2 pts]

It turns out that the eighteen newcomers are merely the first wave of a veritable deluge of enthusiastic arrivals: infinitely many, in fact. ARMLopolis finally decides on a drastic expansion plan: now house numbers will be *rational* numbers. To define $d(p/q)$, with p and q integers such that $pq \neq 0$, write $p/q = 3^k p'/q'$, where neither p' nor q' is divisible by 3 and k is an integer (not necessarily positive); then $d(p/q) = 3^{-k}$.

10a. Compute $d(3/5), d(5/8)$, and $d(7/18)$. [1 pt]

10b. Determine all pairs of relatively prime integers p and q such that $p/q \in \mathcal{N}(4/3)$. [1 pt]

10c. A longtime resident of IMOpia moves to ARMLopolis and hopes to keep his same address. When asked, he says "My old address was e, that is, the sum $\frac{1}{0!} + \frac{1}{1!} + \frac{1}{2!} + \ldots$. I'd really like to keep that address, because the addresses of my friends here are all partial sums of this series: Alice's house is $\frac{1}{0!}$, Bob's house is $\frac{1}{0!} + \frac{1}{1!}$, Carol's house is $\frac{1}{0!} + \frac{1}{1!} + \frac{1}{2!}$, and so on. Just let me know what I have to do in order to be near my friends!" After some head-scratching, the ARMLopolitan planning council announces that this request cannot be satisfied: there is no number, rational or otherwise, that corresponds to the (infinite) sum, or that is arbitrarily close to the houses in this sequence. Prove that the council is correct (and not just bureaucratic). [2 pts]